

- Q.2. a) Implement the **Hill Cipher** in Python using matrix operations. Your program should:
- Accept a key matrix
  - Perform encryption of a plaintext message
  - Perform decryption using the inverse matrix
- b) Implement the **Diffie–Hellman Key Exchange** algorithm in Python to:
- Generate public and private keys
  - Compute and verify the shared secret key between two parties

- a) Implement the **Rail Fence Cipher** in Python. Your program should:
- *Perform encryption using a specified number of rails*
  - *Perform decryption to reconstruct the original message*
- b) Implement the **Diffie–Hellman Key Exchange** algorithm in Python to:
- *Generate public and private keys*
  - *Compute and verify the shared secret key between two parties*

- Q.1. a) Implement the **Playfair Cipher** in Python. Your program should:
- Construct the key matrix
  - Perform encryption of a plaintext message
  - Perform decryption to retrieve the original message
- b) Implement the **RSA Algorithm** in Python, including:
- Key generation (public and private keys)
  - Encryption of a message
  - Decryption to recover the original plaintext

- Q.4. a) Implement the **Rail Fence Cipher** in Python. Your program should:
- Perform encryption using a specified number of rails
  - Perform decryption to reconstruct the original message
- b) Implement the **Diffie–Hellman Key Exchange** algorithm in Python to:
- Generate public and private keys
  - Compute and verify the shared secret key between two parties

Q.3. a) Implement the **Vigenère Cipher** in Python. Your program should:

- Accept a keyword
- Perform encryption of a plaintext message
- Perform decryption to obtain the original text

b) Implement the **RSA Algorithm** in Python, including:

- Key generation (public and private keys)
- Encryption of a message
- Decryption to recover the original plaintext

- Q.3. a) Implement the **Vigenère Cipher** in Python. Your program should:
- Accept a keyword
  - Perform encryption of a plaintext message
  - Perform decryption to obtain the original text
- b) Implement the **RSA Algorithm** in Python, including:
- Key generation (public and private keys)
  - Encryption of a message
  - Decryption to recover the original plaintext

- Q.2. a) Implement the **Hill Cipher** in Python using matrix operations. Your program should:
- Accept a key matrix
  - Perform encryption of a plaintext message
  - Perform decryption using the inverse matrix
- b) Implement the **Diffie–Hellman Key Exchange** algorithm in Python to:
- Generate public and private keys
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- Q.1. a) Implement the **Playfair Cipher** in Python. Your program should:
- Construct the key matrix
  - Perform encryption of a plaintext message
  - Perform decryption to retrieve the original message
- b) Implement the **RSA Algorithm** in Python, including:
- Key generation (public and private keys)
  - Encryption of a message
  - Decryption to recover the original plaintext